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%Aditi Jain(UI24EC02)
clc
clear all
close all
t=linspace(-10,10,201);
subplot(5,1,1);
x1=zeros([1,length(t)]);
for i=1:length(t)
if t(i)==0
x1(i)=1;
break;
else
continue;
end
end
plot(t,x1,"b",linewidth=3)
xlabel("t");
ylabel("x1(t)");
title("t vs x1(t)");
subplot(5,1,2);
x2=zeros([1,length(t)]);
for idx=i:length(t)
x2(idx)=1;
end
plot(t,x2,"r",linewidth=3);
xlabel("t");
ylabel("x2(t)");
title("t vs x2(t)");
subplot(5,1,3);
x3=zeros([1,length(t)]);
for idx=i:length(t)
x3(idx)=1;
end
plot(t,x3,"y",linewidth=3);
xlabel("t");
ylabel("x3(t)");
title("t vs x3(t)");
subplot(5,1,4);
x4=zeros([1,length(t)]);
for idx=i:length(t)
x4(idx)=(t(idx)*t(idx))/2;
end
plot(t,x4,"k",linewidth=3);
xlabel("t");
ylabel("x4(t)");
title("t vs x4(t)");

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subplot(5,1,5);
t1=linspace(-10,0,100);
t2=linspace(0,10,101);
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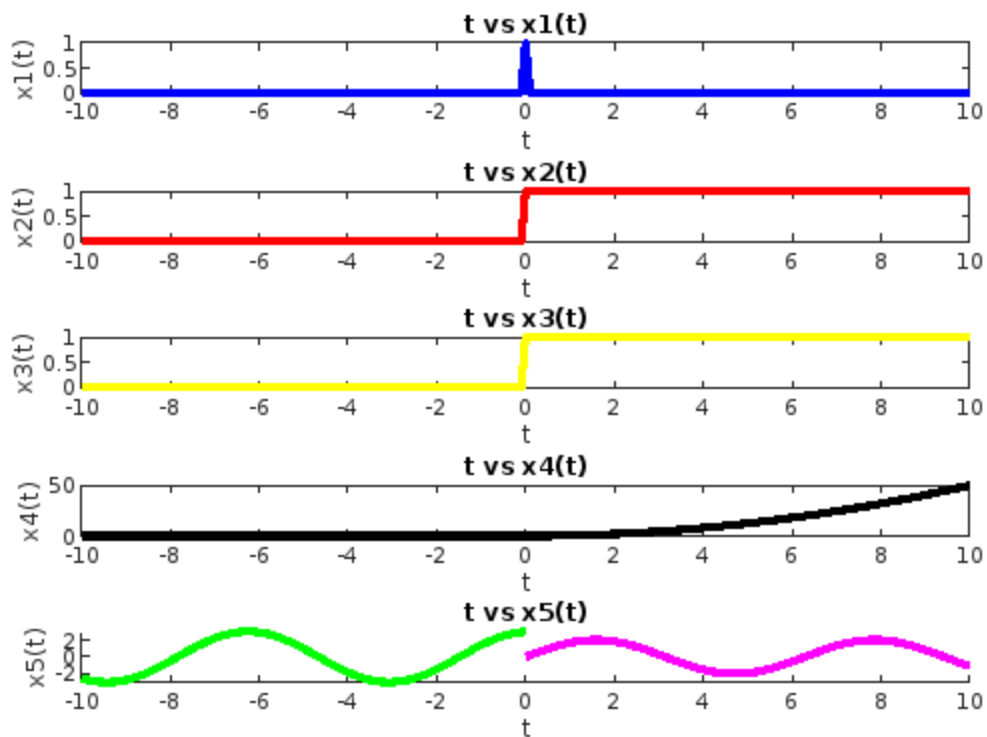
x51=zeros([1,(length(t)/2-0.5)]);
x52=zeros([1,(length(t)/2+0.5)]);
for idx=1:length(t)
z=t(idx);
if idx<i
x51(idx)=3*cos(z);
else
x52(idx-100)=2*sin(z);
end

end
hold on
plot(t1,x51,"g",linewidth=3);
plot(t2,x52,"m",linewidth=3);
hold off
xlabel("t");
ylabel("x5(t)");
title("t vs x5(t)");

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*ans* =

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